

Classical Mechanics

Lecture 6: Constrained Systems and Hamiltonian Motion

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Chapter 1

Constrained Systems and Hamiltonian Motion

Overview. Two constrained systems show why Lagrangian mechanics is so effective: once suitable coordinates have been chosen, the equations of motion and their conservation laws follow without an explicit calculation of the constraint forces. The same examples then lead to Hamiltonian mechanics, in which coordinates and momenta form phase space, the harmonic oscillator becomes a rigid rotation, and Hamilton's two first-order equations replace each second-order Euler–Lagrange equation.

1.1 What It Means to Solve a Mechanical Problem

The immediate purpose of the Lagrangian method is practical. A wedge sliding on a floor and a double pendulum can both be treated with Newton's laws, but the unknown contact forces and tensions quickly make the bookkeeping unpleasant. Once the rules of Lagrangian mechanics are known, the same problems become a sequence of controlled steps:

1. choose independent coordinates that label an instantaneous configuration;
2. express the kinetic and potential energies in those coordinates;
3. form $L = T - V$ and apply the Euler–Lagrange equations.

For the present purpose, a problem is already *solved* when its equations of motion have been written down correctly. One may still want a closed-form trajectory, but that is a separate question. Nonlinear differential equations can be integrated numerically once the mechanical setup is complete. The hard part is often not the integration; it is finding variables that incorporate the constraints without introducing redundant unknown forces.

1.2 A Particle Sliding on a Moving Wedge

Consider a wedge of mass M constrained to slide without friction along a horizontal plane. A point particle of mass m is constrained to remain on the inclined face. There are two degrees of freedom: one tells us where the wedge is, and one tells us where the particle is on the wedge.

Let

$$\begin{aligned} X &= \text{horizontal position of the wedge corner,} \\ x &= \text{horizontal displacement of the particle from that corner.} \end{aligned} \quad (1.1)$$

The particle's absolute horizontal coordinate is therefore $X + x$. The choice is not unique. We could instead use distance along the slope or the particle's absolute horizontal coordinate. Any nonredundant pair that labels the same configurations is legitimate, though some pairs expose the structure more clearly than others.

For a fixed slope angle β , the vertical displacement y is not an independent coordinate:

$$y = x \tan \beta, \quad \dot{y} = \dot{x} \tan \beta. \quad (1.2)$$

We specialize to $\beta = 45^\circ$, so that $y = x$ and $\dot{y} = \dot{x}$. This simplification removes trigonometric factors without changing the point of the example.

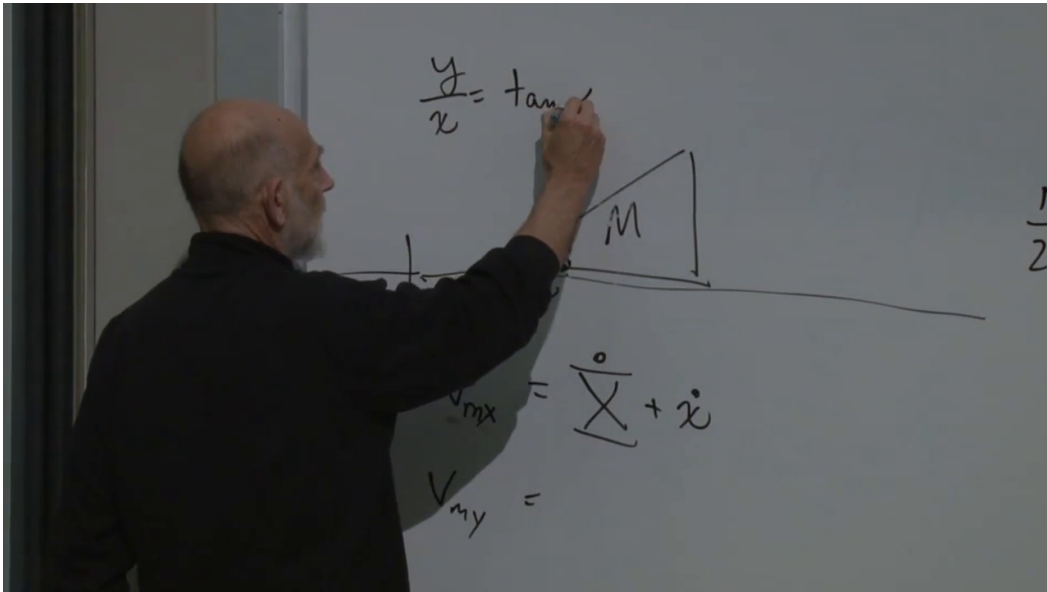


Figure 1.1: The moving wedge, the constrained particle, and the horizontal and vertical velocity components used in the 45° example.

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The wedge has kinetic energy

$$T_M = \frac{M}{2} \dot{X}^2. \quad (1.3)$$

The particle's horizontal velocity is $\dot{X} + \dot{x}$, while its vertical velocity is $\dot{x}$. Its kinetic energy is consequently

$$T_m = \frac{m}{2} \left[(\dot{X} + \dot{x})^2 + \dot{x}^2 \right]. \quad (1.4)$$

Taking the zero of gravitational potential at $x = 0$, we have $V = mgx$. Thus

$$L = \frac{M}{2} \dot{X}^2 + \frac{m}{2} \left[(\dot{X} + \dot{x})^2 + \dot{x}^2 \right] - mgx. \quad (1.5)$$

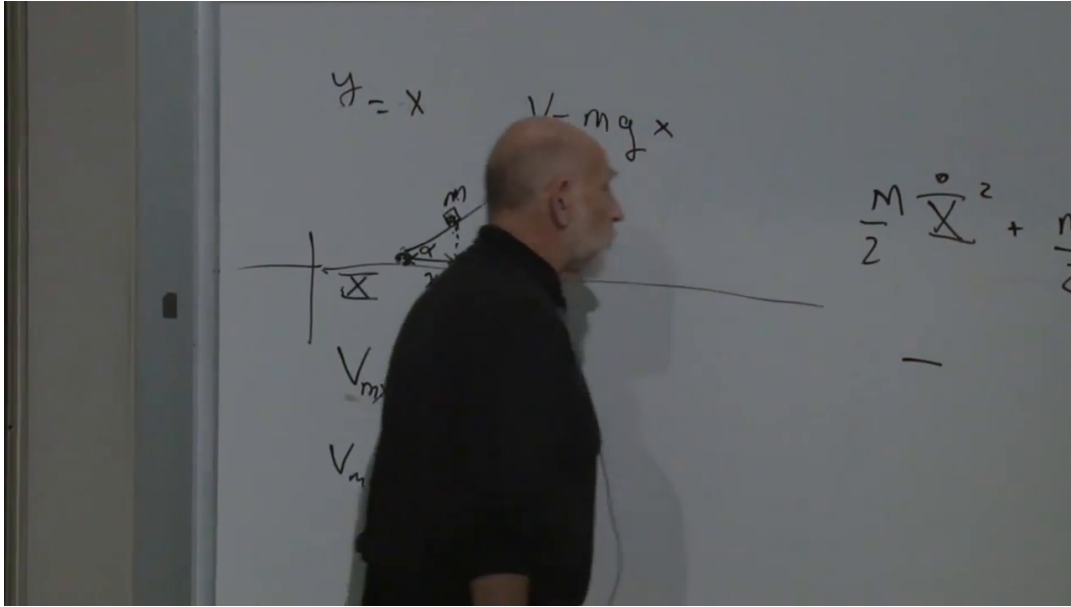


Figure 1.2: The wedge geometry, $V = mgx$, and the two kinetic-energy contributions assembled on the board.

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1.2.1 Canonical Momentum Is Defined by the Lagrangian

The canonical momenta are defined by the Lagrangian, not by a guess based on a single particle velocity:

$$p_X = \frac{\partial L}{\partial \dot{X}} = M\dot{X} + m(\dot{X} + \dot{x}), \quad (1.6)$$

$$p_x = \frac{\partial L}{\partial \dot{x}} = m(\dot{X} + \dot{x}) + m\dot{x}. \quad (1.7)$$

The first expression is the ordinary total horizontal momentum: the wedge contributes $M\dot{X}$, and the particle contributes $m(\dot{X} + \dot{x})$. It appears here as the momentum conjugate to the coordinate X .

The Lagrangian has no explicit dependence on X . A horizontal translation of the whole apparatus acts as

$$\delta X = \epsilon, \quad \delta x = 0. \quad (1.8)$$

The general infinitesimal symmetry rule

$$\delta q_i = \epsilon f_i(q), \quad Q = \sum_i p_i f_i \quad (1.9)$$

therefore gives $Q = p_X$. Equivalently, X is cyclic and

$$\dot{p}_X = \frac{\partial L}{\partial X} = 0. \quad (1.10)$$

There is no corresponding vertical translation symmetry. The wedge is not allowed to move vertically, and changing the particle's height changes the gravitational potential. Horizontal momentum is conserved; vertical momentum of this constrained subsystem is not.

$$P_X = \frac{\partial L}{\partial \dot{X}} = M \dot{X} + m(\dot{X} + \dot{x})$$

$$P_x = m(\dot{X} + \dot{x}) + m \dot{x}$$

Figure 1.3: The canonical momenta conjugate to the wedge coordinate X and the relative particle coordinate x .

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For the relative coordinate, the Euler–Lagrange equation is

$$\dot{p}_x = \frac{\partial L}{\partial x} = -mg. \quad (1.11)$$

This line required a sign correction at the board. The dependable route is to begin with $L = T - V$: since $V = mgx$, the derivative of the Lagrangian is $-mg$. The fact that x was introduced as a horizontal displacement does not remove gravity from its equation, because the constraint ties that horizontal displacement to the vertical height $y = x$.

No normal force or reaction force has appeared. Such forces are whatever they must be to maintain the constraints. Newton’s method would introduce them and solve for them along with the accelerations; the generalized coordinates have made them unnecessary.

1.3 The Double Pendulum and the Art of Coordinates

Now suspend two unit masses from two massless rods of unit length. The first rod makes an angle θ with the downward vertical. For the second degree of freedom we have a choice. We could use a second absolute angle ϕ , or we could use the relative angle α between the rods.

The relative angle is the more revealing choice. Temporarily turn gravity off. A rigid rotation of the complete apparatus changes the absolute orientation but not the angle between the rods:

$$\theta \longrightarrow \theta + \epsilon, \quad \alpha \longrightarrow \alpha. \quad (1.12)$$

With this choice the rotational symmetry acts on one coordinate only. This does not make other coordinate systems wrong. It makes the symmetry, and therefore the conserved quantity, easier to see.

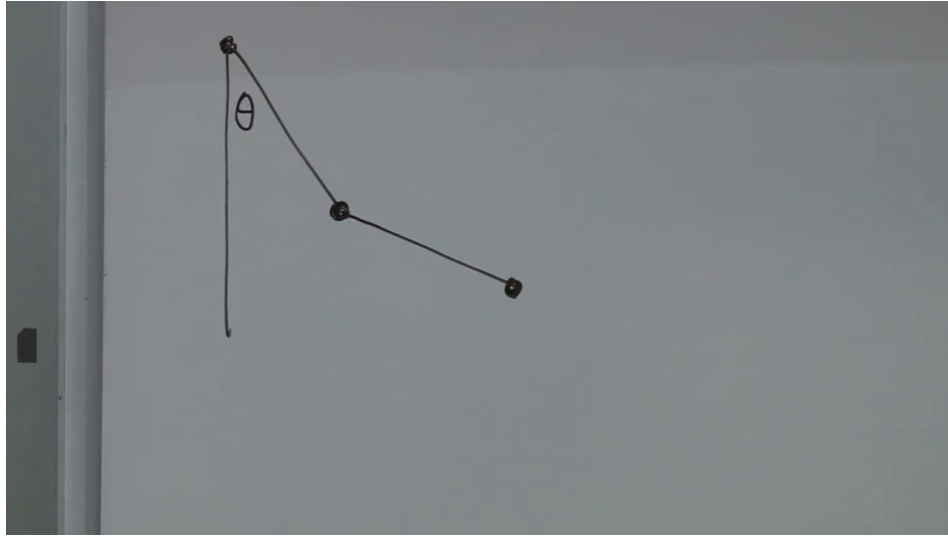


Figure 1.4: The double pendulum with θ as the first absolute angle and α as the second rod's angle relative to the first.

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To obtain the kinetic energy, use Cartesian coordinates as temporary bookkeeping devices. With the vertical coordinate measured downward, the first bob is at

$$x_1 = \sin \theta, \quad y_1 = \cos \theta, \quad (1.13)$$

and hence

$$\dot{x}_1 = \dot{\theta} \cos \theta, \quad \dot{y}_1 = -\dot{\theta} \sin \theta, \quad \dot{x}_1^2 + \dot{y}_1^2 = \dot{\theta}^2. \quad (1.14)$$

The second bob is at

$$x_2 = \sin \theta + \sin(\theta + \alpha), \quad (1.15)$$

$$y_2 = \cos \theta + \cos(\theta + \alpha), \quad (1.16)$$

so, for example,

$$\dot{x}_2 = \dot{\theta} \cos \theta + (\dot{\theta} + \dot{\alpha}) \cos(\theta + \alpha), \quad (1.17)$$

with an analogous expression for $\dot{y}_2$.

Adding the two kinetic energies and using $\cos A \cos B + \sin A \sin B = \cos(A - B)$ gives the consistent result

$$T = \dot{\theta}^2 + \frac{1}{2}(\dot{\theta} + \dot{\alpha})^2 + \dot{\theta}(\dot{\theta} + \dot{\alpha}) \cos \alpha. \quad (1.18)$$

Question from the audience. Does this expression already include the kinetic energy of both bobs, and has a factor of one half been lost? ^a

Response. It includes both. It comes from $\frac{1}{2}(\dot{x}_1^2 + \dot{y}_1^2 + \dot{x}_2^2 + \dot{y}_2^2)$. Two equal contributions combine to produce the first term $\dot{\theta}^2$; no factor is missing.

^aLecture timestamp: 00:38:06–00:38:27.

Without gravity, $L = T$ has no explicit θ . Thus

$$\dot{p}_\theta = \frac{\partial L}{\partial \theta} = 0, \quad (1.19)$$

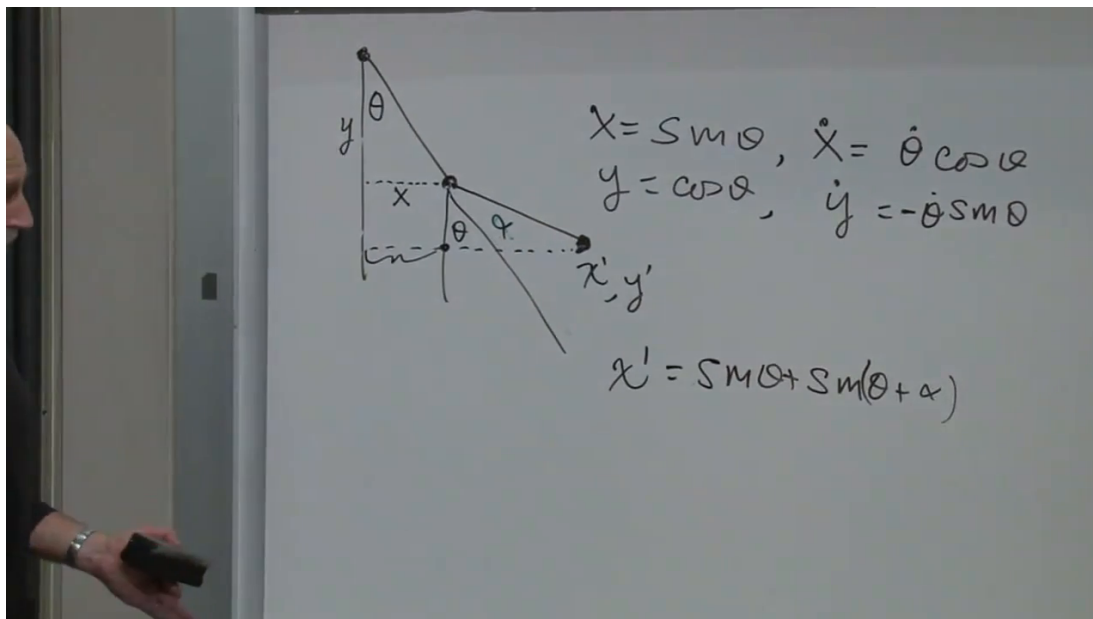


Figure 1.5: Auxiliary Cartesian coordinates for the two bobs. They are used to compute the velocities, not introduced as additional degrees of freedom.

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and p_θ is the total angular momentum. Notice the precise meaning of the statement that θ does not appear: $\dot{\theta}$ certainly appears, but the undifferentiated coordinate does not.

Restoring gravity breaks the global rotational symmetry. For the downward vertical convention above, one convenient choice of zero gives

$$V = -g(y_1 + y_2) = -g[2 \cos \theta + \cos(\theta + \alpha)]. \quad (1.20)$$

An additive constant would change nothing. The important point is that θ now appears explicitly, so its conjugate momentum is no longer conserved. Energy remains conserved because the Lagrangian has no explicit time dependence.

Question from the audience. Should the board's $\dot{\theta} - \dot{\alpha}$ be $\dot{\theta} + \dot{\alpha}$? ^a

Response. For the orientation used in Eqs. (1.16)–(1.18), it must be the plus sign. Reversing the definition of positive α changes the corresponding signs, but every occurrence must then be changed consistently. The board mixed the two conventions.

^aLecture timestamp: 00:43:00–00:43:54.

Once T and V are known, the remaining equations are obtained by differentiation. Rod tensions and the forces that enforce fixed lengths have again disappeared into the choice of coordinates.

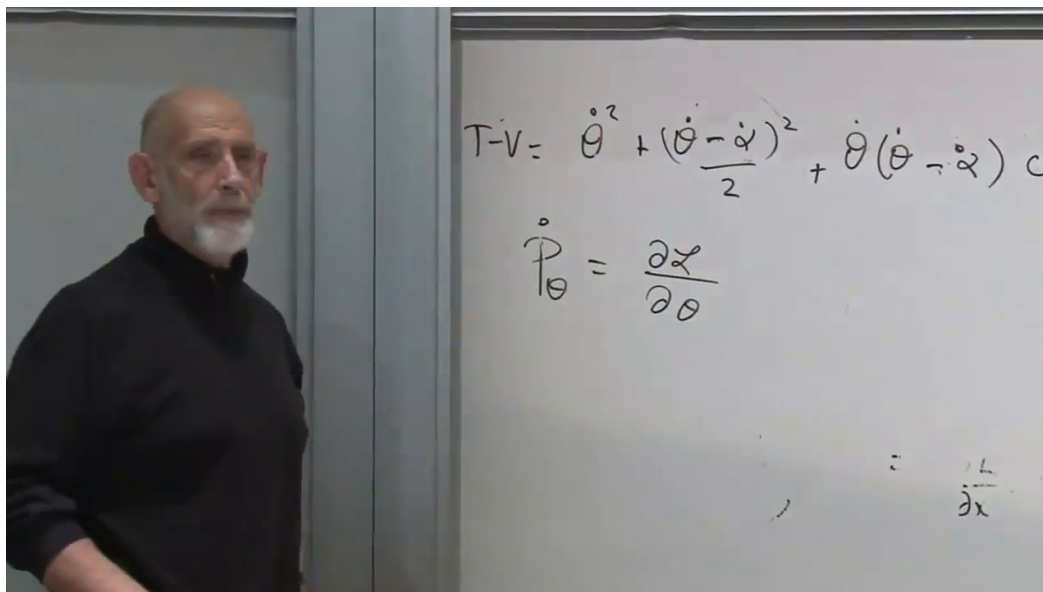


Figure 1.6: The no-gravity pendulum calculation and $\dot{p}_{\theta} = \partial L / \partial \theta$. The upper board line preserves the sign-inconsistent local notation discussed in the question; the normalized formula is Eq. (1.18).

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1.3.1 What conservation does and does not imply

Question from the audience. Can one see directly from the equations whether the double pendulum is chaotic? ^a

Response. Usually not by simple inspection. As parameters or energies vary, a system may pass from regular to chaotic behavior. The subject requires more structure than the raw appearance of the equations, and it will be taken up later.

^aLecture timestamp: 00:44:57–00:45:25.

Question from the audience. In a conservative system, is every $\dot{p}_i$ equal to zero? ^a

Response. No. The general equation is $\dot{p}_i = \partial L / \partial q_i$. It vanishes only when q_i is cyclic. A cyclic coordinate is the especially simple case in which a symmetry has been made to shift one coordinate while leaving the others fixed.

^aLecture timestamp: 00:45:25–00:46:24.

If the only objective is numerical evolution, an inconvenient coordinate system still works; it merely produces longer equations. The reward for a clever system is conceptual: the conserved momentum is visible before the equations are solved.

Question from the audience. Without gravity, does a double pendulum behave as simply as a single free pendulum? ^a

Response. A single free rotor turns uniformly because its one angular momentum fixes its motion. A double pendulum can execute very complicated relative motion while preserving the total angular momentum. If both angular velocities initially vanish it remains at rest; otherwise the two parts interact.

^aLecture timestamp: 00:47:02–00:47:35.

Question from the audience. Are $\dot{\theta}$ and $\dot{\alpha}$ individually constant when gravity is absent? ^a

Response. No. Conservation says $\dot{p}_\theta = 0$, not $\ddot{\theta} = 0$ or $\ddot{\alpha} = 0$. The expression for $p_\theta = \partial L / \partial \dot{\theta}$ contains both angular velocities and depends on α . The rods transmit torques, allowing the two bobs to exchange angular momentum and energy. This is analogous to two masses joined by a spring: total momentum and center-of-mass velocity are constant, while the individual velocities oscillate. The rotational analogue is a particular collective combination, not either angle by itself.

^aLecture timestamp: 00:47:36–00:52:42.

Question from the audience. Are the forces that exchange angular momentum carried through the rods? ^a

Response. Yes. A force transmitted along a rod need not point through the pivot of the other motion, so it can exert a torque. A torque is simply the rotational tendency of a force, and here it produces the nonzero individual angular accelerations.

^aLecture timestamp: 00:52:43–00:53:34.

1.4 The Hamiltonian as a Third Formulation

The Hamiltonian is energy, but that is only its first significance. It is also the basis of a third formulation of mechanics. Newton's equations use forces and accelerations. The principle of stationary action gives the Euler–Lagrange equations. Hamiltonian mechanics is equivalent to the latter for regular systems, but its variables and geometry look different.

Instead of q_i and $\dot{q}_i$, use q_i and their conjugate momenta p_i . A state of a system with n degrees of freedom requires $2n$ numbers in either description: positions and velocities, or positions and momenta. The bridge between them is

$$p_i = \frac{\partial L}{\partial \dot{q}_i}. \quad (1.21)$$

When these relations can be inverted for the velocities $\dot{q}_i(q, p)$, define

$$H(q, p) = \sum_i p_i \dot{q}_i - L(q, \dot{q}). \quad (1.22)$$

The velocities on the right are then eliminated in favor of q and p . This invertibility is the regular Legendre-transform assumption. Singular Lagrangians require the more general theory of constraints; they are not handled by this elementary construction.

1.4.1 Reversibility before calculus

Recall a finite collection of abstract states connected by arrows. A deterministic rule may still fail to be reversible if two earlier states flow into one later state. The future is unique, but the past is not. Reversing all the arrows gives the opposite defect: one state branches into several futures. In the first picture arrows converge; in the second they diverge.

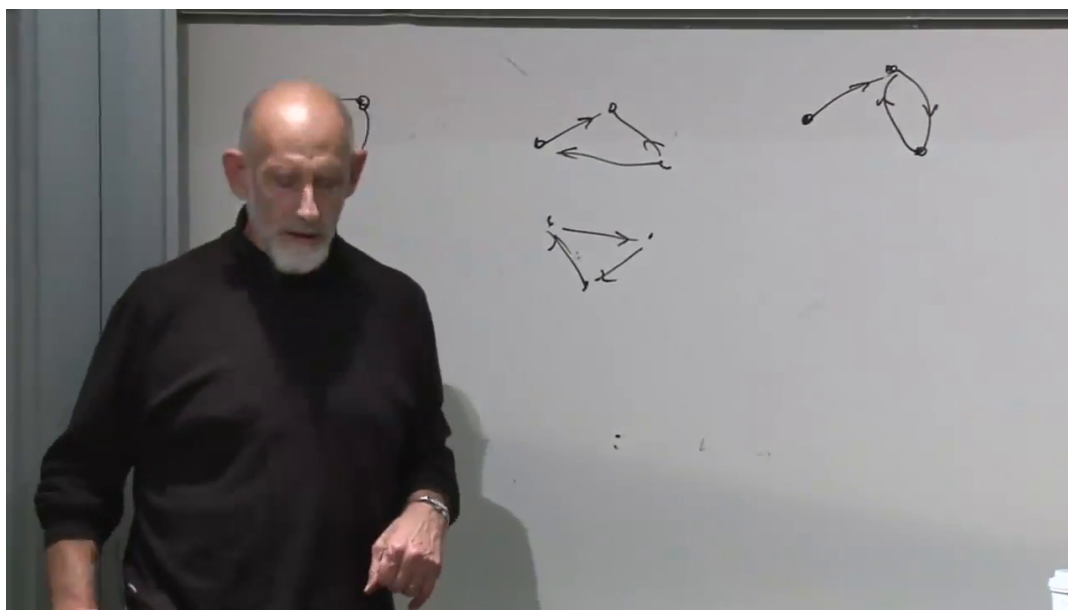


Figure 1.7: Abstract state maps used to contrast reversible cycles with forbidden convergence and divergence.

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Hamiltonian flow will supply the continuous version of this statement. A cloud of phase-space points may move and deform, but it cannot acquire a sink that destroys distinctions between states or a source that manufactures several futures from one state. The proof is deferred; for now the diagrams motivate the geometry.

Question from the audience. Should the abstract state diagram be interpreted as an object moving in a gravitational field? ^a

Response. No. The dots are merely abstract states and the arrows specify a rule for changing state. No spatial or gravitational meaning has been assigned to the drawing.

^aLecture timestamp: 00:58:02–00:58:35.

Question from the audience. How is a forbidden convergent state map distinguished from a black hole? ^a

Response. Only with difficulty and subtlety. It is a fair question, but a correct answer requires a careful account of the complete state and its quantum-gravitational evolution. The elementary diagram is not enough to settle it.

^aLecture timestamp: 00:58:40–00:59:28.

1.5 An Oscillator Adapted to Phase Space

Before deriving the general equations, consider a harmonic oscillator written in deliberately adapted variables:

$$L = \frac{1}{2\omega} \dot{q}^2 - \frac{\omega}{2} q^2. \quad (1.23)$$

Its conjugate momentum is

$$p = \frac{\partial L}{\partial \dot{q}} = \frac{\dot{q}}{\omega}, \quad \dot{q} = \omega p. \quad (1.24)$$

Substitution in Eq. (1.22) gives

$$H = p\dot{q} - L \quad (1.25)$$

$$= \omega p^2 - \frac{\omega p^2}{2} + \frac{\omega q^2}{2} = \frac{\omega}{2} (p^2 + q^2). \quad (1.26)$$

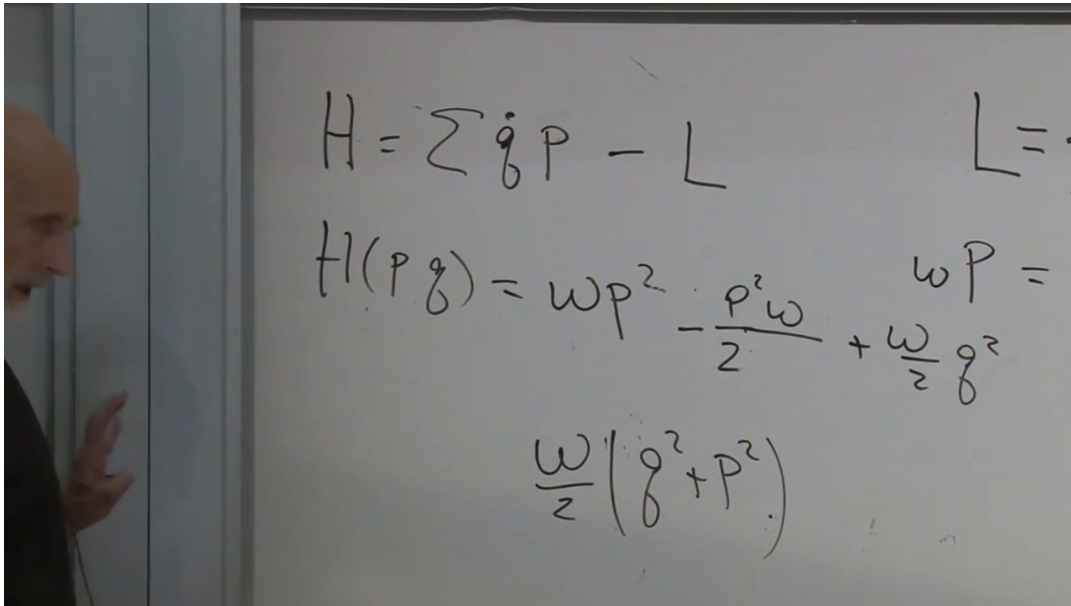


Figure 1.8: The Legendre transform of the adapted oscillator and the symmetric Hamiltonian $H = \frac{\omega}{2} (p^2 + q^2)$.

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The space of q and p is phase space. One degree of freedom gives a two-dimensional plane; two degrees of freedom give a four-dimensional phase space. At fixed oscillator energy,

$$p^2 + q^2 = \frac{2H}{\omega}, \quad (1.27)$$

so the trajectory remains on a circle. Energy conservation confines the motion to one such curve.

1.6 Deriving Hamilton's Equations

The derivation is an exercise in the differential of a function of several variables. For $F(q, p)$,

$$\delta F = \sum_i \left(\frac{\partial F}{\partial q_i} \delta q_i + \frac{\partial F}{\partial p_i} \delta p_i \right). \quad (1.28)$$

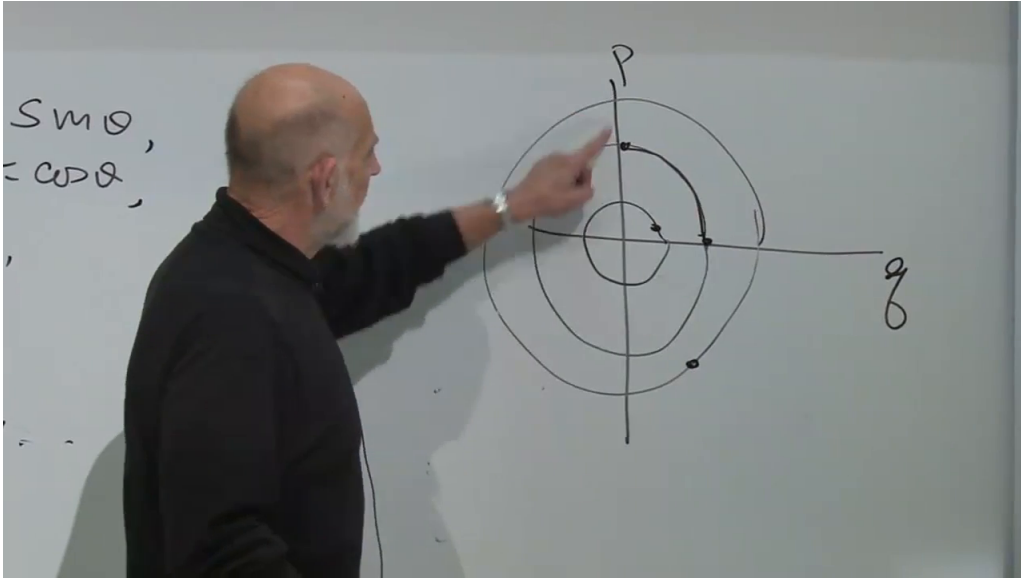


Figure 1.9: Nested constant-energy circles in the oscillator phase plane, with q horizontal and p vertical.

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The coefficients can be read in either direction: the coefficient of δq_i is $\partial F/\partial q_i$, and the coefficient of δp_i is $\partial F/\partial p_i$.

Now vary the defining expression for H :

$$\delta H = \sum_i (\dot{q}_i \delta p_i + p_i \delta \dot{q}_i) - \delta L \quad (1.29)$$

$$= \sum_i \left(\dot{q}_i \delta p_i + p_i \delta \dot{q}_i - \frac{\partial L}{\partial q_i} \delta q_i - \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i \right). \quad (1.30)$$

Because $\partial L/\partial \dot{q}_i = p_i$, the two $\delta \dot{q}_i$ terms cancel. The Euler–Lagrange equations give $\partial L/\partial q_i = \dot{p}_i$, leaving

$$\delta H = \sum_i (\dot{q}_i \delta p_i - \dot{p}_i \delta q_i). \quad (1.31)$$

Comparing Eq. (1.31) with the general differential in Eq. (1.28), the independent coefficients give Hamilton’s equations:

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}.} \quad (1.32)$$

For n coordinates these are $2n$ first-order equations. The Euler–Lagrange formulation instead supplies n second-order equations. The amount of initial data and the physical content are identical.

1.7 Oscillator Motion as a Rigid Rotation

Applying Eq. (1.32) to Eq. (1.26) gives

$$\dot{q} = \omega p, \quad \dot{p} = -\omega q. \quad (1.33)$$

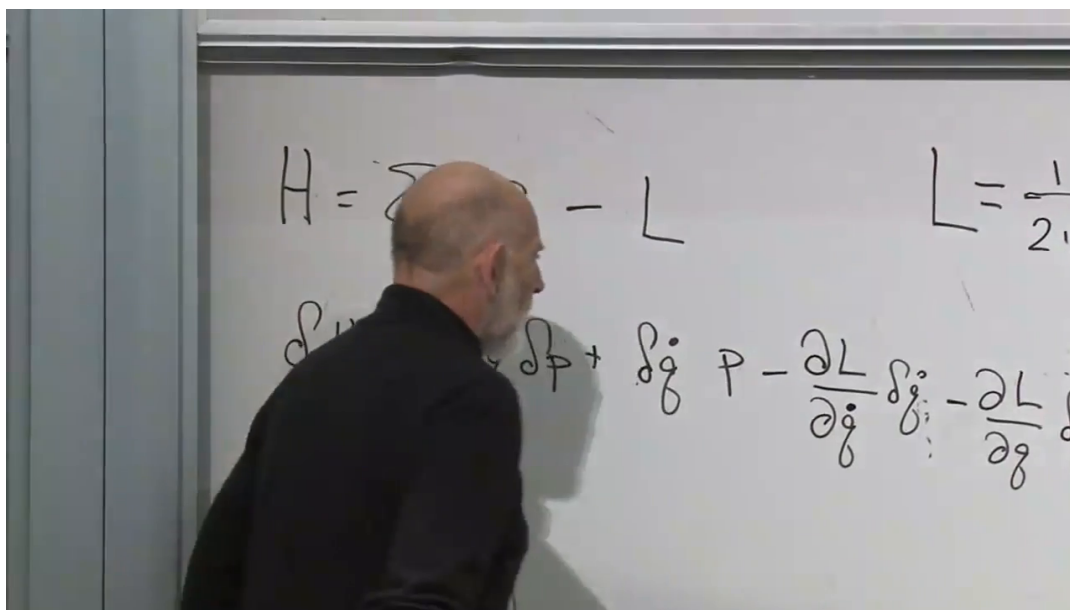


Figure 1.10: The product-rule expansion of δH before the $p_i \delta \dot{q}_i$ terms cancel.

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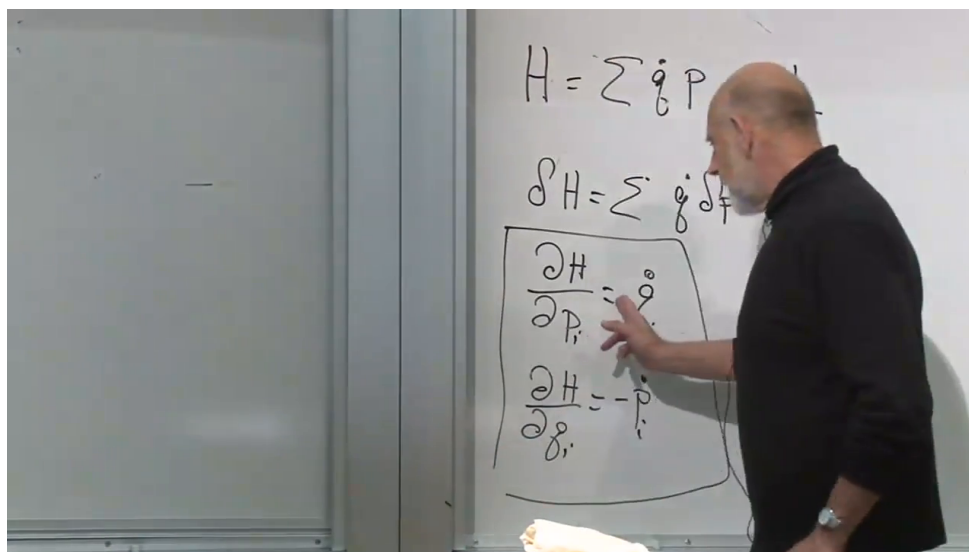


Figure 1.11: The Hamiltonian definition, its variation, and the boxed canonical equations.

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Differentiating the first equation and using the second,

$$\ddot{q} = \omega \dot{p} = -\omega^2 q. \quad (1.34)$$

Thus the two first-order equations are exactly the usual second-order oscillator equation, together with the definition of momentum.

In phase space the velocity vector at (q, p) is $(\dot{q}, \dot{p}) = (\omega p, -\omega q)$, tangent to the energy circle. Every radius rotates with the same angular speed ω . The entire plane may be imagined as a rigid pinwheel or as an incompressible fictitious fluid. Its points circulate; they do not squeeze into a sink or fly apart from a source.

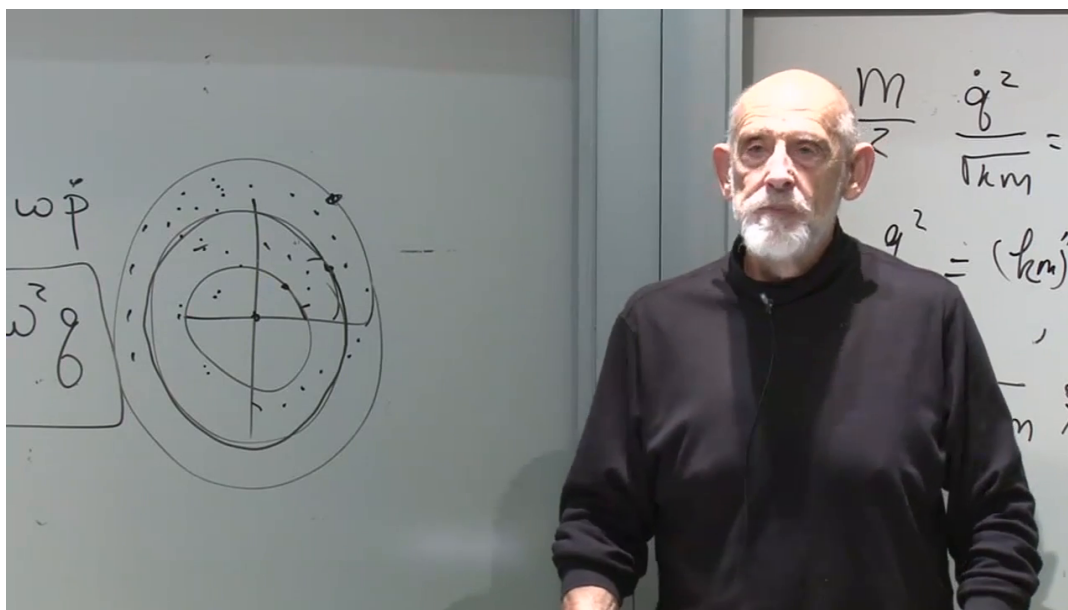


Figure 1.12: A cloud of initial points distributed over nested phase-space trajectories. Hamiltonian evolution moves the cloud rather than collapsing it into a preferred state.

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The next step will be to turn the fluid analogy into a theorem: Hamiltonian flow has no local convergence or divergence. That result reconnects the continuous mechanics with the reversible state maps considered at the beginning of the course.

1.8 Questions That Expose the Coordinate Choice

Question from the audience. Is the assumption that velocities can be solved in terms of q and p a theorem or an empirical property of physical systems? ^a

Response. Within this elementary development it is an empirical regularity, with a deeper motivation from quantum mechanics. More precisely, Eq. (1.21) must have a nonsingular Jacobian with respect to the velocities. Modern mechanics also contains important singular Lagrangians, especially constrained and gauge systems, but their Hamiltonian formulation needs additional constraint machinery.

^aLecture timestamp: 01:29:08–01:29:45.

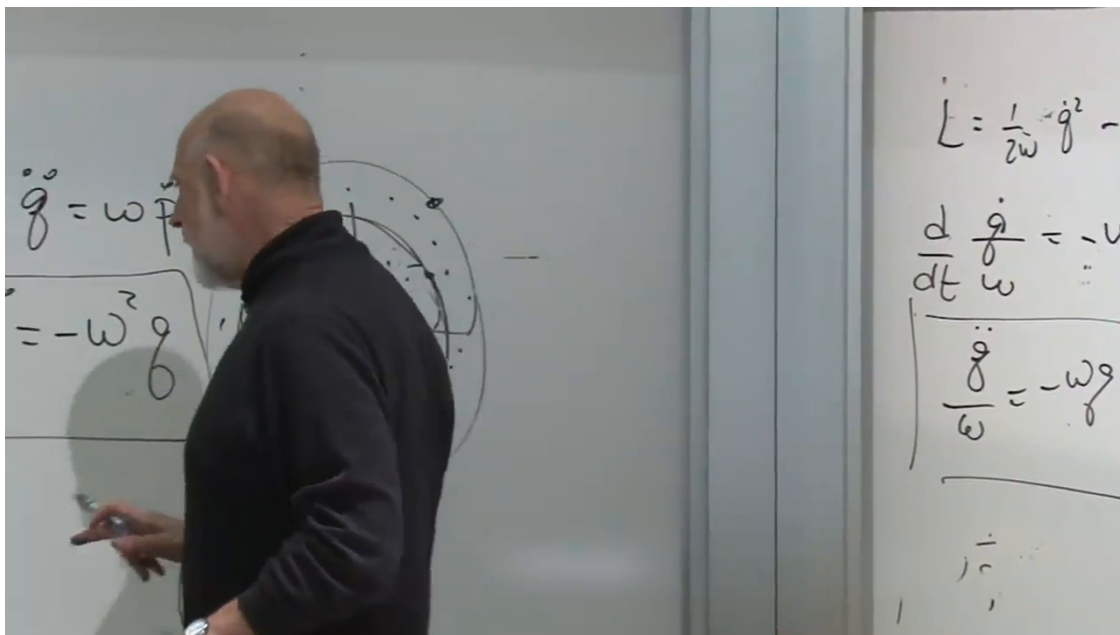


Figure 1.13: The oscillator phase portrait beside the equivalent Euler–Lagrange equation $\ddot{q}/\omega = -\omega q$.

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An objection about units explains why the oscillator was written in the unfamiliar form of Eq. (1.23). Begin instead with the ordinary coordinate x :

$$L = \frac{m}{2} \dot{x}^2 - \frac{k}{2} x^2, \quad \omega = \sqrt{\frac{k}{m}}. \quad (1.35)$$

Choose

$$q = (km)^{1/4} x. \quad (1.36)$$

Then $q^2 = \sqrt{km} x^2$ and $\dot{q}^2 = \sqrt{km} \dot{x}^2$, which converts Eq. (1.35) precisely into Eq. (1.23). The new q is an adapted generalized coordinate, not a length, and its conjugate p is not the ordinary mechanical momentum $p_x = m\dot{x}$.

Question from the audience. How can $\omega p = \dot{q}$ be dimensionally consistent, and why were these variables chosen? ^a

Response. The symbol q is not required to have dimensions of length, and its conjugate p is not required to have the dimensions of $m\dot{x}$. The rescaling in Eq. (1.36) preserves the dimensions of the Lagrangian and makes q and p enter H symmetrically. In the original variables,

$$H = \frac{p_x^2}{2m} + \frac{kx^2}{2},$$

so constant-energy curves in the (x, p_x) plane are ellipses. In the adapted (q, p) plane they are circles. The rescaling was chosen to make that symmetry visible.

^aLecture timestamp: 01:29:46–01:39:22.

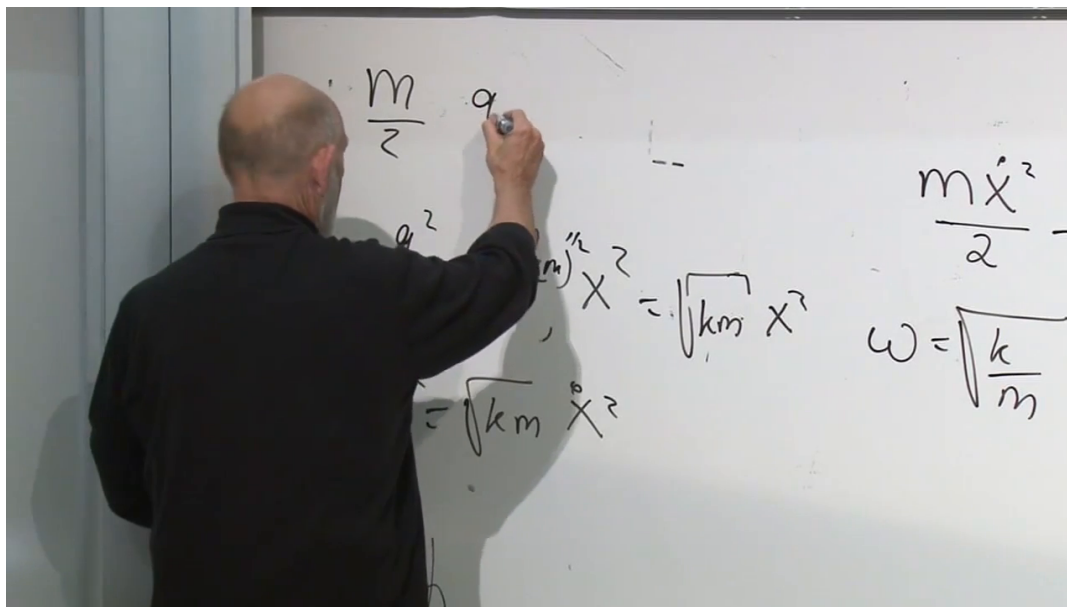


Figure 1.14: The dimensional rescaling from the usual oscillator coordinate x to the adapted coordinate $q = (km)^{1/4}x$.

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Question from the audience. How can $2n$ first-order Hamilton equations be equivalent to n second-order Lagrange equations? ^a

Response. For the oscillator, differentiate $\dot{q} = \omega p$ and substitute $\dot{p} = -\omega q$; the pair becomes $\ddot{q} = -\omega^2 q$. Starting directly from the Lagrangian gives

$$\frac{d}{dt} \left(\frac{\dot{q}}{\omega} \right) = -\omega q,$$

which is the same second-order equation. One Hamilton equation can be viewed as defining momentum from velocity, while the other evolves that momentum.

^aLecture timestamp: 01:39:24–01:42:30.

1.9 Why Phase Space and Potential Energy Matter

Question from the audience. Does the power of the Hamiltonian mean that phase space is a better geometrical representation than configuration space? ^a

Response. They serve different purposes. Phase space makes cycles, constant-energy surfaces, and the absence of convergence or divergence unusually transparent. Configuration space may be better for other questions. The deeper reason for emphasizing H is that it becomes a central and more intuitive object in quantum mechanics, one of the modern subjects toward which this mechanics is directed.

^aLecture timestamp: 01:42:33–01:44:38.

Question from the audience. Should potential energy be regarded as the integral of force? ^a

Response. For conservative forces that statement is valid, up to a constant and with the appropriate sign. The more useful direction is to regard the potential as primary and derive the generalized force from it:

$$F_i = -\frac{\partial V}{\partial q_i}.$$

This is exactly the term that appears in $\dot{p}_i = \partial L / \partial q_i$ when $L = T - V$ and the kinetic energy has no explicit dependence on q_i .

^aLecture timestamp: 01:44:40–01:45:45.

Question from the audience. Which frontier problems is this path toward modern physics meant to illuminate? ^a

Response. Quantum gravity, cosmology, elementary-particle physics, the Higgs sector, neutrinos and relativity, string theory, and quantum field theory all lie along that path. The enduring theoretical problem is to fit the different pieces into one coherent structure, especially quantum mechanics with gravity and cosmology with particle physics. This was a statement of the frontier as it appeared in 2011, not a claim that its solution was already known.

^aLecture timestamp: 01:45:46–01:47:47.

The two constrained examples and the oscillator now fit into one picture. Generalized coordinates remove unnecessary forces. Symmetry selects conserved canonical momenta. The Legendre transform exchanges velocities for momenta, and Hamilton's equations turn dynamics into a flow on phase space. For the oscillator that flow is a rigid rotation; for a general system its geometry will provide the precise continuous meaning of reversible motion.